Aerogel supercapacitors for energy storage and electronics applications

Aerogel Supercapacitors are promising energy storage devices which can store electrical energy higher than that of conventional capacitors and are ideal choice for many advanced applications. C-MET has developed complete indigenous technology for the preparation of carbon aerogel and carbon aerogel based Aerogel Supercapacitor cells up to 50 F. The Aerogel Supercapacitor technology developed at C-MET is protected by Indian patents.



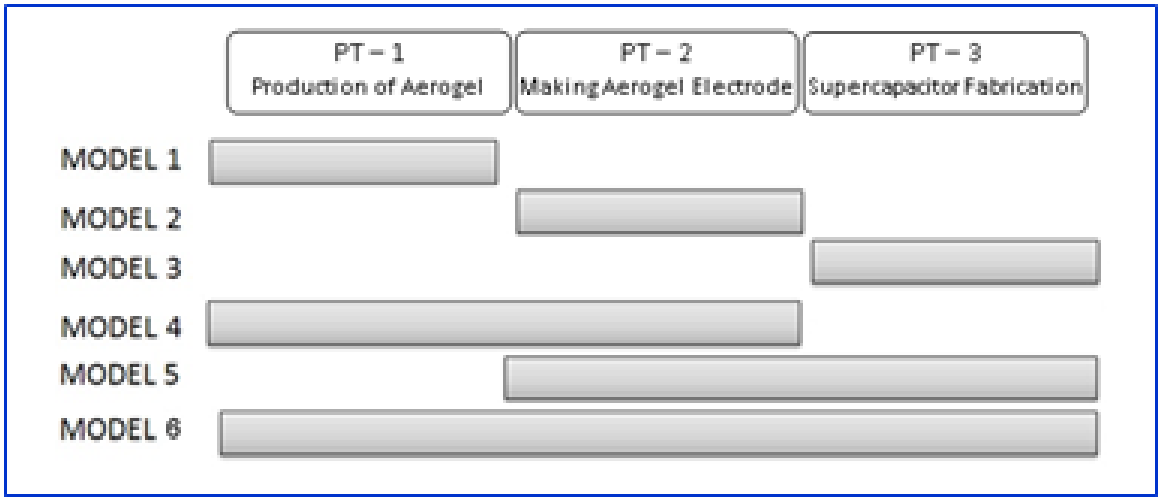
Fig 1. Photograph of Aerogel Supercapacitor developed at C-MET

2. Main uses and domain

Supercapacitors are of high demand for Consumer Electronics sector and finds applications in devices like DVD, Emergency lighting, Digital Camera, Mobile phones, Toys, PC, e-book readers etc. In addition to this, Supercapacitors also find applications in Industrial, Automotive and Renewable Energy sectors.

3. Features and Technical Specifications

C-MET Thrissur has established facilities for fabrication of Aerogel supercapacitor, which consists of three sub-technologies which are available for the ToT (Transfer of Technology) either in total or in part/combinations as descried below



Technology transfer models

PT-1: Technology for preparation of 'Carbon Aerogel'

PT-2: Technology for making 'Aerogel Electrode '

PT-3: Technology for fabrication of 'Aerogel Supercapacitor'

Specifications

(a) Carbon aerogel

BET Specific Surface area	: 800-1000 m^2/g
Average Pore diameter	: 2-5 nm
Specific Capacitance	: 50-55 F/g

(b) Carbon aerogel based super capacitor electrode

Width	: 25-30 mm
Thickness	: 200-350 µm.

(c) Aerogel Supercapacitor

Cell type	: Cylindrical with leads
Cell capacitance	: 25 F
Capacitance tolerance	: ± 15%
Cell Operating voltage	: 2.5 ~2.7 V DC

During training, Aerogel supercapacitor technology will be demonstrated at C-MET Thrissur for the following batch size

i) Carbon aerogel preparation : Batch size- 0.5 Kg

ii) Aerogel Electrode Fabrication : Electrodes required for fabrication of 50 number of 25 F aerogel supercapacitor (Single electrode Length 25-32cm and Width 2.5-3.0 cm)

iii) Aerogel Supercapacitor fabrication : 50 nos. of cylindrical cells of 25±10% F

4. Technology Transfer Fee

Sl. No.	TOT for different stages	ToT cost (Rs. in lakhs) (Excluding applicable taxes)
1.	PT-1	7.79
2.	PT -2	7.23
3.	PT -3	7.89
4.	PT- 1 & 2	12.61
5.	PT– 2 & 3	12.75
6.	PT – 1, 2 & 3	20.61

5. Contact details for Techno-commercial Information

The Director

Centre for Materials for Electronics Technology (C-MET),

M.G. Kavu, Athani (P.O.), Thrissur - 680 581, Kerala.

Tel: 0487-2201757; e-mail: raghu@cmet.gov.in, cmett@cmet.gov.in